is connected to an A.C Voltage V1, a flux will set up in the core and e.m.f E1 will be induced in the winding. Such induced e.m.f is taken by tapping at point C. There are N2 turns between B and C, an e.m.f E2 is induced B and C.

When load is connected across the terminals B and C a current I2 flows. The m.m.f due to I2 will be balanced by m.m.f due to I1. This arrangement is referred as an Auto transformer. If point C is sliding contact, a continuously variable output voltage can be made available.

So,

$$\frac{V_2}{V_1} = \frac{I_1}{I_2} = \frac{N_2}{N_1} = K$$

Advantages

1. Copper required is very less.
2. The efficiency is higher.
3. Required less conducting material and hence cost is less.
4. More smooth & continuous variation of voltage.

Disadvantages

1. Direct link between high voltage and low voltage sides there is no isolation as in the case of a two winding transformer.
2. The short circuit current is greater than that of a two winding transformer.

6.5.4 Applications of Auto Transformer

1. As a booster of supply voltage to a small extent.
2. Auto transformers are used to start the induction motor.
3. It can be used to vary the voltage to the load, smoothly from zero to the rated voltage.

6.6 LOSSES IN A TRANSFORMER

The losses in a transformer consists of I^2R loss (or) copper loss and iron loss (or) core loss.

I^2R Loss (or) Copper Loss

These losses occur in primary and secondary windings. Copper loss in a transformer is a variable loss. It varies as the square of the load current (From short circuit test this can be determined).

Iron Loss (or) Core Loss

Iron loss consists of hysteresis and eddy current losses. They occur in the transformer core due to the alternating flux (from open circuit test, this can be determined.)

Hysteresis Loss

When the iron core is subjected to an alternating flux hysteresis loss takes place.

Eddy Current Loss

Eddy current is induced in the cores. This loss is due to the flow of eddy current. Thin laminations are used to reduce the eddy current loss.

6.6.1 Efficiency of a Transformer

The efficiency of a transformer is the ratio of output power to input power.

EMA_English_Chp06.indd 96 11/9/2020 3:19:36 PM